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Direct dating of paleomagnetic results from Precambrian sediments in the Amazon craton: Evidence for Grenvillian emplacement of exotic crust in SE Appalachians of North America

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Abstract

We apply a new diagenetic dating technique to determine the age of magnetization for Precambrian sedimentary rocks in the SW Amazon craton. Two new paleomagnetic poles are reported from the rocks of the Aguapeí Gp.: red beds of the Fortuna Fm. (P_{lat}=59.8°N, P_{lon}=155.9°E, A_{95}=9.5, K=14, 18 sites, N/n 128/115, Q=5) and the reverse-polarity mudstones of the overlying Vale da Promissão Formation (P_{lat}=49.5°N, P_{lon}=89.3°E, A_{95}=12.5, K=30, 6 sites, N/n=94/80, Q=4). The Fortuna Fm. magnetization is hosted by massive, interstitial hematite cement and constitutes a post-depositional remanence. The age of diagenesis of the red beds is well-constrained by the 1149±7 Ma U–Pb age of authigenic xenotime rims on detrital zircons determined by SHRIMP analysis. The magnetite-hosted remanence of the Vale da Promissão Fm. may be detrital in origin, but the age of deposition is poorly constrained. The reliable and precisely-dated Fortuna Fm. paleomagnetic pole fixes the paleogeographic position of the Amazon craton near the SE Appalachians portion of North America at 1.15 Ga. These data demonstrate a mobile Grenvillian link between these two cratons, and support the recent identification of Amazon crust in the Blue Ridge province region of North America.
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1. Introduction

Reliable paleomagnetic data provide a quantitative, global reference frame for shifting tectonic plates. For Pre-Jurassic times, this reference frame is indispensable due to the destruction of geological evidence such as marine magnetic anomalies or the non-unique aspects of faunal correlations or paleoclimate indicators. The "reliability" is established by linking two observations: reconstruction of the time-averaged field from an isolated magnetic remanence and the determination of the age of the magnetization. Improvements in instrumentation and magnetic cleaning techniques now allow for routine measurement of a wide range of magnetic remanences (10^{-3}–10^{2} A/m) from diverse rocks. In contrast, the age of magnetization is established by relative dating methods or is assumed to date from the time of rock formation, which is especially problematic for barren Precambrian sedimentary rocks where the rock age is not easily determined. A survey of the global database shows that sedimentary rocks constitute more than one-third of

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the >1500 paleomagnetic poles reported for the Precambrian, with an average uncertainty >10% of the rock age. To put this in perspective, a widely adopted criterion rejects magnetizations with >4% age uncertainty (Van der Voo, 1990), eliminating >80% of extant results from Precambrian sedimentary rocks. A new technique for the dating of diagenesis holds great potential for providing precise age constraints hitherto lacking in paleomagnetic investigations of Precambrian sediments.

A key issue in Precambrian geology revolves around the supercontinent Rodinia (Torsvik, 2003), the configuration and position of which are considered to have had a major role in the onset of Neoproterozoic global glaciations (Donnadieu et al., 2004). The link between the Amazon craton and eastern Laurentia across the Grenvillian collision zone (1.2–1.0 Ga) is a key feature of the Rodinia supercontinent, the rupture of which marks the opening of the Iapetus Ocean at the beginning of the Phanerozoic era. The configuration of the two continents in the Precambrian supercontinent varies in the original reconstructions following the match of either collision zone metamorphism (Hoffman, 1991) or post-rift thermal subsidence and stratigraphy (Bond et al., 1984; Dalziel, 1992). Data from the Amazon craton itself suggest a different configuration beginning with a southerly collision site (Tohver et al., 2002) followed by strike-slip migration of the Amazon towards a more central location near the Grenville Province of Ontario–New York (Tohver et al., 2005a).

The recent identification of exotic Amazonian crust lodged within the basement of the southeastern Appalachians (Carrigan et al., 2003; Loewy et al., 2003; Tohver et al., 2004a) appears to corroborate this strike-slip passage, but only well-dated paleomagnetic data can provide the necessary spatial reference frame to confirm this scenario [e.g., Hanson et al. (2004)].

In this paper we present new paleomagnetic and geochronological data from late Mesoproterozoic sediments of the Aguapeí Group, western Brazil. These sediments were deposited in the southwestern margin of the Amazon craton, over the Paragua shield. Barren sedimentary rocks from the Precambrian era are difficult to date, especially so for clastic rocks. Paleomagnetic studies of these rocks are further complicated by the uncertain origin of the magnetization as either a depositional remanence (DRM) or post-depositional remanence formed by the growth of an authigenic phase (CRM) (Verosub, 1977). We overcome these problems by using a novel technique to date diagenesis, U–Pb analysis of authigenic xenotime rims on detrital zircon. Together, these new data provide a high quality, well-dated paleomagnetic pole that constrains the

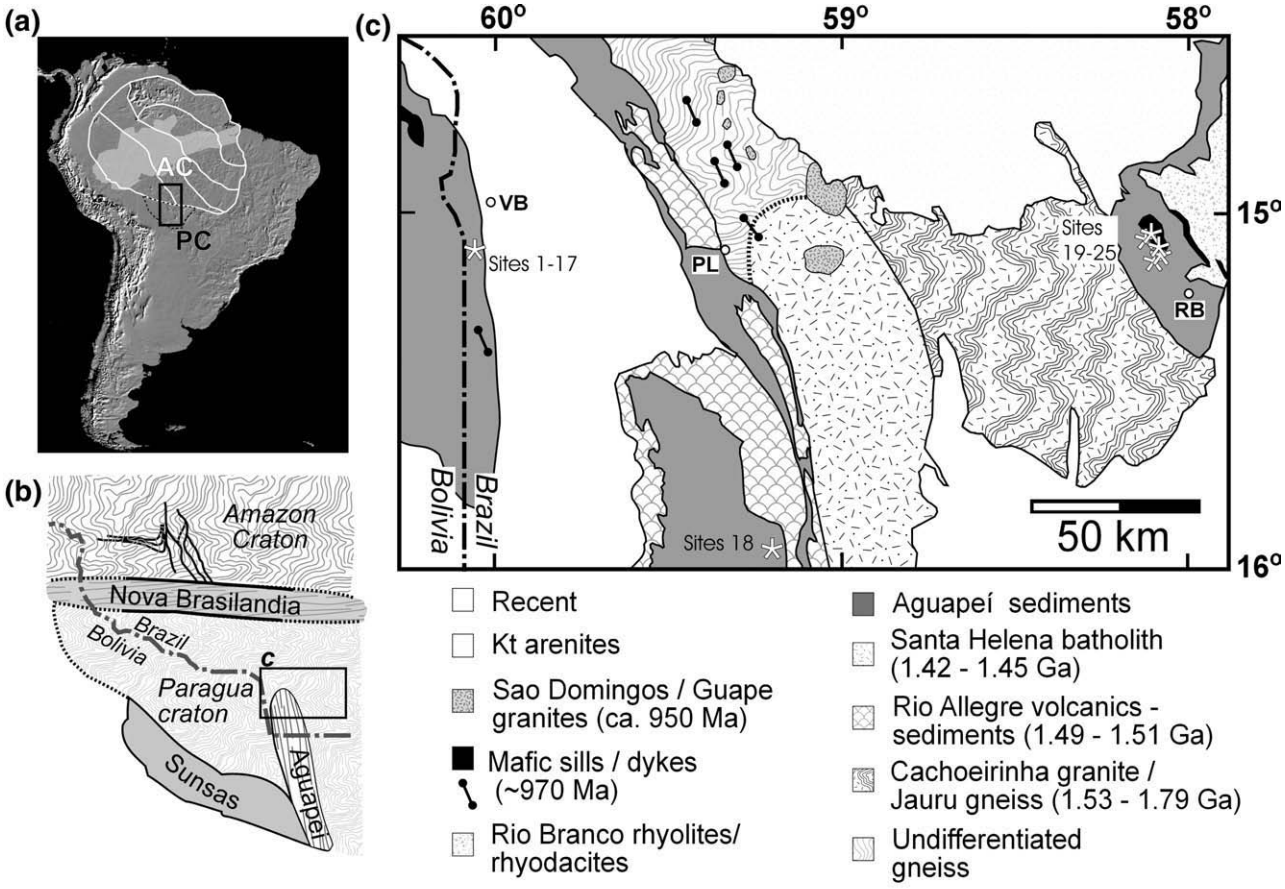


Fig. 1. a) Inset: South American location map with outline of geochronological provinces of the Amazon Craton (AC) adapted from Santos et al. (2000) showing also the Paragua Craton (PC) and the localization of the study area. Light grey indicates the ~E–W trending Paleozoic Basin. b) General geometry of SW Amazon and Paragua cratons showing the (ca. 1.1–1.0 Ga) Nova Brasilândia, Aguapeí, and Sunsas belts. c) Geological map of the southwestern portion of the Amazon craton, after Geraldes et al. (2001) and Ruiz (2005) showing the location of sampled units of the Aguapeí Group (sites indicated by stars). VB — Vila Bela; RB — Rio Branco, PL — Pontes e Lacerda.

paleogeographic position of the Amazon craton during the Grenvillian interval (1.2–1.0 Ga), with important implications for the evolution of the Rodinia supercontinent and for models of the Grenville collision belt.

1.1. SW Amazon Regional Geology

The Amazon Craton (Fig. 1) is bisected by the E–W trending Paleozoic basin (Solimões-Amazonas) that underlies the Amazon River. A late Archean nucleus in the NE portion of the craton is bordered by accretionary belts that are progressively younger to the SW (Santos et al., 2000; Tassinari et al., 2000), although the general lack of outcrop and geophysical data has led to debate over the position and significance of the various geochronological provinces that comprise the craton (Pinho et al., 2003). The youngest addition to the Amazon craton is the Paragua block, a subcontinental shield in eastern Bolivia (Fig. 1) that has been interpreted as either separate from, or as a subprovince of the Amazon craton before 1.1 Ga (Boger et al., 2005). A minimum age for a joint Amazon–Paragua history derives from the 1.1 Ga high-grade metamorphism of the suturing Nova Brasilândia Metasedimentary Belt (Santos et al., 2000; Rizzotto, 1999). Collision must have taken place before metamorphism, given the clockwise pressure–temperature path of continental collisions; e.g., the 50–55 Ma collision between India and Eurasia caused Himalayan regional metamorphism 20–40 Ma later (Hodges, 2000). Because rates of orthogonal convergence and burial are related (Thompson et al., 1997), we infer a 50 Ma lag time for the transpressive Nova Brasilândia belt (Rizzotto, 1999) and a common geography for the Paragua–Amazon craton by 1.15 Ga.

Rocks of the correlated Sunsás and Aguapeí Gps. overlie the 1.7–1.4 Ga Paragua basement rocks of Bolivia and Brazil (Fig. 1), respectively. In Bolivia, the deposition of ca. 6 km Sunsás Gp. psammites and pelites is ascribed to a deltaic-fluvial environment (Litherland et al., 1989). Deformation and granite intrusion during the eponymous orogeny at ≥1076 Ma (Boger et al., 2005) were marked by prominent NW–SE trending mylonite belts with sinistral displacement (Litherland et al., 1989). In Brazil, the 1–2 km thick Aguapeí Gp. comprises basal conglomerates and red sandstones (Fortuna Fm.), overlain by pelitic rocks and fine lithic, sub-arkosic rocks (Vale da Promissão Fm.), topped by quartz sandstones of the Morro Cristalina Fm. (Souza and Hildred, 1980; Saes and Leite, 1993). Detrital zircons from the basal Fortuna Fm. range in age from 1453±10 Ma to 1165±27 Ma (n=89) (Santos et al., 2001; Leite

Table 1
Paleomagnetic results from the Aguapeí sediments

Site	Coordinates	N/n	Site mean direction				VGP	
	Lat. (°S)/long. (°W)		Dec. (°)	Inc. (°)	α_{95}	k	P_{lon} (°E)	P_{lat} (°N)
1	14° 55.721′/60° 01.197′	5/5	6.6	−38.0	15.7	25	76.5	81.0
2	14° 55.721′/60° 01.197′	5/5	358.0	−63.1	16.9	21	122.8	60.3
3	14° 55.721′/60° 01.197′	4/4	339.4	−55.8	7.7	141	156.8	61.8
4	14° 55.721′/60° 01.197′	9/7	302.5	−60.7	11.9	27	169.4	34.0
5	14° 55.721′/60° 01.197′	10/10	317.5	−67.8	8.5	33	154.1	40.5
6	14° 55.721′/60° 01.197′	7/7	321.5	−55.2	8.1	56	171.5	49.8
7	14° 55.721′/60° 01.197′	9/9	358.1	−66.3	7.8	44	122.2	56.2
8	14° 55.721′/60° 01.197′	7/7	297.5	−53.1	5.0	148	179.4	30.9
9	14° 55.721′/60° 01.197′	7/7	345.4	−59.6	6.0	102	143.7	61.5
10	14° 55.721′/60° 01.197′	9/6	329.5	−50.6	11.9	104	174.2	57.7
11	14° 55.721′/60° 01.197′	9/8	328.3	−56.4	6.2	81	166.0	54.3
12	14° 55.721′/60° 01.197′	6/6	347.4	−55.5	7.5	82	145.7	66.1
13	14° 55.721′/60° 01.197′	9/9	319.1	−62.3	3.9	174	162.0	45.0
14	14° 55.721′/60° 01.197′	5/5	2.4	−46.9	10.4	55	110.8	76.6
15	14° 55.721′/60° 01.197′	6/6	19.8	−44.9	12.0	32	65.2	68.2
16	14° 55.721′/60° 01.197′	8/4	331.1	−57.1	18.4	26	162.8	55.8
17	14° 55.721′/60° 01.197′	4/4	303.5	−47.2	13.2	49	185.5	36.3
18	16° 03.556′/59° 13.003′	9/6	1.9	−36.3	7.5	80	97.3	85.5
19	15° 21.212′/58° 05.048′	14/13	357.9	−51.4	7.0	36	128.0	73.2
20	15° 24.131′/58° 05.947′	10/10	195.1	50.4	5.8	69	263.3	−69.1
21	15° 23.996′/58° 05.950′	13/10	215.8	53.6	7.9	38	248.8	−52.8
22	15° 17.992′/58° 02.642′	15/9	232.5	67.6	12.2	25	263.8	−35.2
23	15° 07.838′/58° 07.480′	21/20	221.1	66.9	4.7	49	266.8	−42.1
24	15° 07.838′/58° 07.480′	17/15	209.1	70.8	7.2	29	279.1	−44.1
25	15° 07.838′/58° 07.480′	18/16	192.0	70.3	3.0	150	291.0	−49.7
Mean (upward inclinations)		18[a]	339.0	−56.7	7.3	24	155.9	59.8
							A_{95}=9.5	K=14
Mean (downward inclinations)		6	210.2	64.0	9.3	53	269.3	−49.5
							A_{95}=12.5	K=30

N/n — analyzed specimens/specimens used in mean directions. Dec. — declination; Inc. — inclination, α_{95} — radius of the 95% confidence cone; K — precision parameter; Plon — Paleolongitude; Plat — Paleolatitude.

[a] This mean represents red beds (sites 1 to 18) from the Fortuna Formation (see text).

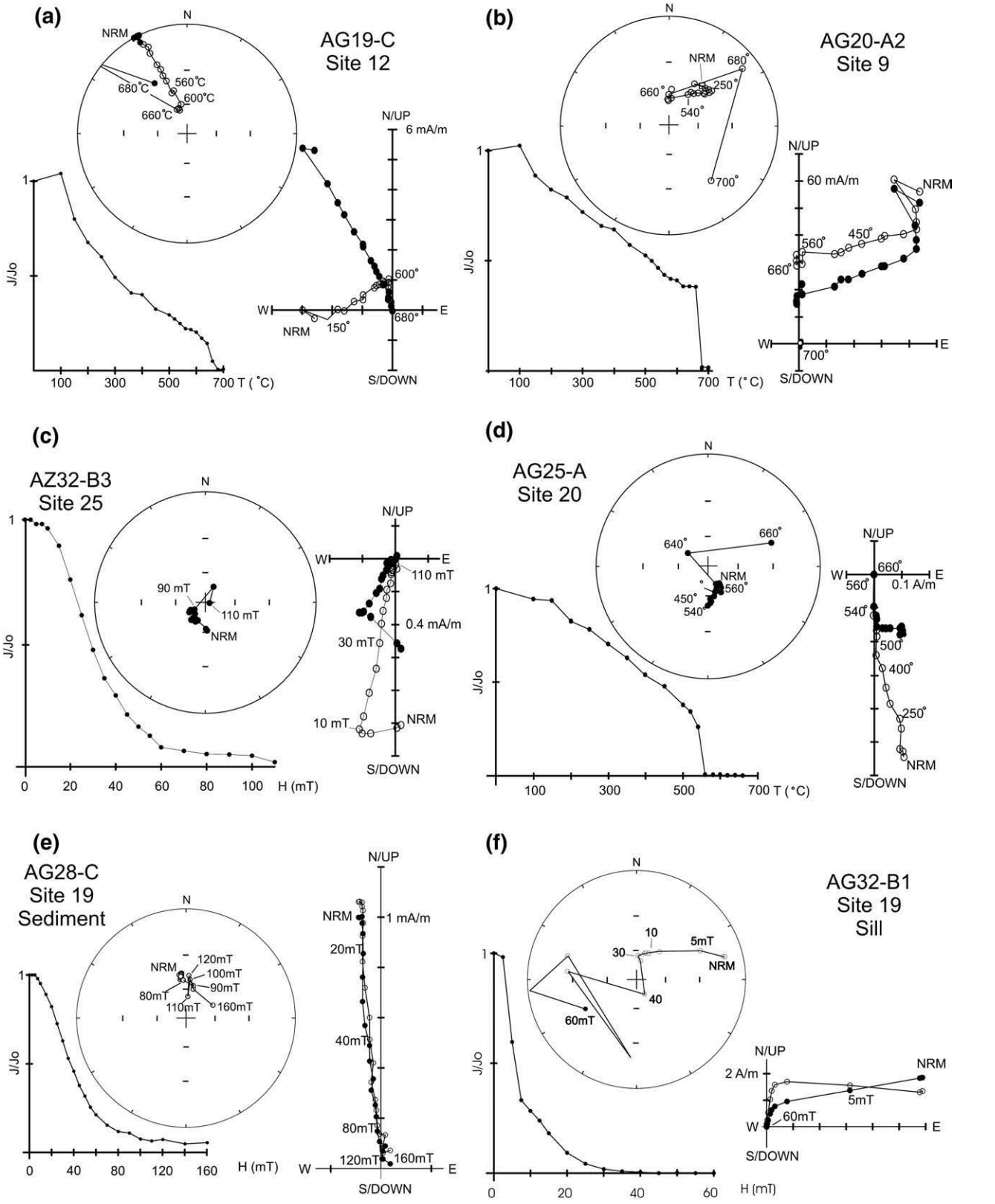


Fig. 2. Typical demagnetization trajectories for red beds from the Fortuna Formation (a, b), Vale da Promissão Formation (c, d), and for samples of site 19 (e — sandstone of Vale da Promissão Formation; f — mafic sill). Magnetization intensity decay curves, stereographic projection, and orthogonal projection are represented for each sample. Open (full) symbols correspond to negative (positive) inclinations in the stereographic projections. Open (full) symbols correspond to vertical (horizontal) projections in orthogonal plots.

and Saes, 2003), a maximum age limit on the Aguapeí Gp deposition. The minimum deposition age is bracketed by deformation of these rocks under lower greenschist conditions at ca. 920 Ma (Geraldes et al., 1997). This deformation is ~100 Ma younger than the more extensive Sunsas deformation to the SSW, indicating that strain in these two belts was not necessarily related. Deformation of the 50–100 km wide Aguapeí belt is bounded to the west along the Brazil–Bolivia frontier and to the east by the Brazilian town of Rio Branco, with both flanks characterized by subhorizontal layers of Aguapeí sediments with minimal metamorphism.

2. Sampling and methods

Paleomagnetic sampling consisted of 123 oriented cores and 7 hand samples collected from 25 sedimentary levels of the Fortuna Formation and Vale da Promissão Formation of the Aguapeí Group (Fig. 1b). Cylindrical cores were drilled using a portable, gasoline-powered rock drill, and whenever possible, both magnetic and solar compasses were used for sample orientation. Details of site locations are included in Table 1. A total of seventeen sites were taken from steep bluffs of the reddish Fortuna Formation near the town of Vila Bela on the Brazil–Bolivia border. This flank is located outside the core of the Aguapeí Belt, with deformation restricted to gentle, open folds (Fig. 1b). The first 15 sites were taken from a continuous, ~22 m thick stratigraphic section, with sites 16 and 17 collected from exposures of uncertain stratigraphic position relative to the first sites. Seven sites (sites 19–25) were collected from the eastern flank of the Aguapeí belt, close to the towns of Rio Branco and Salto do Céu (Fig. 1b). The grey sediments from these sites are from the Vale da Promissão Formation that overlies the Fortuna Formation. The sediments from site 19 (road outcrop) are cross-cut by a sill which appears locally as small blocks of drift. Unfortunately, it was not possible to collect samples from sediments far from the sill for a baked contact test. This, and other basic sills were also collected for paleomagnetic analysis and their results have been presented by Elming et al. (submitted). One of the sills was selected for dating, and the $^{40}Ar/^{39}Ar$ analysis on biotite yielded a well defined plateau age of 980.6±1.7 Ma. Only one site (site 18 in Table 1) was collected from the internal, more deformed portion of the Aguapeí belt just to the south of the town of Pontes e Lacerda (Fig. 1b), although the beds at this site were flat-lying.

Samples were cut into 2.2×2.5 cm cylindrical specimens and submitted to step-wise thermal and alternating field (AF) demagnetization in the paleomagnetic laboratories at the Universidade de São Paulo (USP), Brazil, and the University of Luleå, Sweden. Polished thin sections were made from representative samples, with reflected light microscopy and scanning electron microscopy in conjunction with energy dispersive spectroscopy used to examine the opaque mineralogy. A total of 236 specimens from 25 sedimentary layers were analyzed after magnetic cleaning. Magnetic remanence was measured using a 2 G cryogenic magnetometer after step-wise treatment by alternating fields (AF) or thermal demagnetization. AF demagnetization was carried out using an automated three-axis AF-demagnetizer (upper limit of 170 mT) and thermal demagnetization was carried out in a Magnetic Measurements (MMTD60) oven or a Schonsted oven, with peak temperatures within ±2 °C and a heating time of 1 h. In São Paulo these instruments are housed in a magnetically shielded room with ambient field <1000 nT. Magnetic components for each specimen were identified in orthogonal plots, and calculated using least-squares fits (Kirschvink, 1980). Fisherian statistics (Fisher, 1953) were used to calculate vector mean directions and paleomagnetic poles. Additional magnetic measurements were performed in a Moslpin vibrating sample magnetometer for hysteresis curves, and a Magnetic Measurements pulse magnetizer was used for isothermal remanence acquisition.

Zircons were separated from a sample of the Fortuna Formation (JO2), the basal member of the Aguapeí Group, using standard techniques: crushing, sieving, milling, separation by Frantz magnetic separator, and concentration on heavy liquids. Approximately 2500 grains were mounted in epoxy, polished, and carbon coated for scattered electron microscopy (SEM). Grains were examined using backscattered electron imagery for the presence of authigenic xenotime overgrowths wide enough for SHRIMP analysis (≥10 μm). SHRIMP analysis was carried out at the Curtin University of Technology, Western Australia, using an analytical spot of about 10 μm and six scans. All data are corrected for matrix effect using the MG1 standard (490 Ma; (Fletcher et al., 2004)), and are quoted at 2σ level. Presented ages are $^{207}Pb/^{206}Pb$ ages corrected for common lead using the ^{204}Pb content.

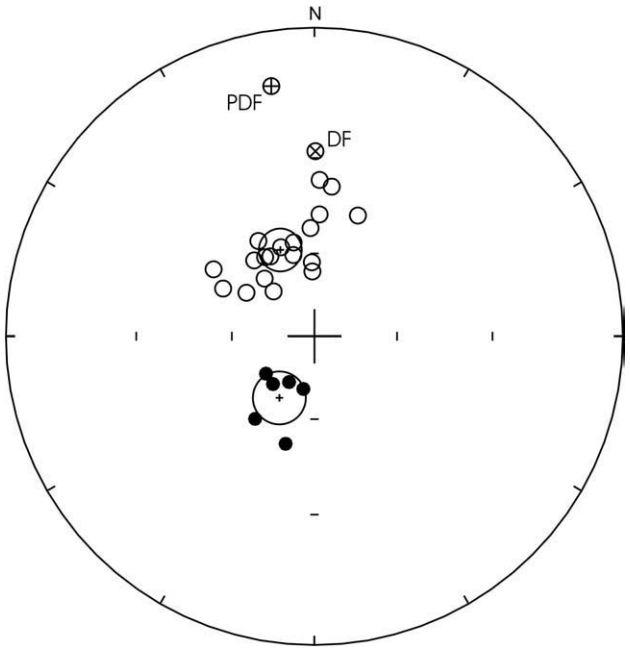


Fig. 3. Site mean directions for the Aguapeí sediments: normal directions are from the Fortuna Formation, and reverse directions are from the Vale da Promissão Formation. Mean characteristic directions for each formation are represented by crosses within 95% confidence circles (α_{95}). Open (full) symbols correspond to negative (positive) inclinations in the stereographic projections. PDF: present-day geomagnetic field; DF: dipolar field.

3. Results and observations

3.1. Magnetic components

The natural remanent magnetization (NRM) intensities vary from 0.224×10^{-3} A/m up to values as high as 17.4 A/m. AF demagnetization on the red sediments of the Fortuna Formation eliminates a soft component (7.5–30 mT) that persists during thermal demagnetization to temperatures up to 560 °C–580 °C (Fig. 2a, b). This overprint having been removed, we observe a northerly, upward direction that persists to temperatures between 600 °C and 660 °C (Fig. 2a to b). As for the grey sandstones and pelites of the Vale da Promissão Formation, most samples record a soft, randomly-oriented magnetic component with coercivities from 0 to 20–30 mT (Fig. 2c) and unblocking temperatures up to 400–450 °C (Fig. 2d). Removal of this overprint reveals a south/southwestern, relatively steep downward inclination direction with higher coercivities and unblocking temperatures that range from 450° to 580 °C (Fig. 2c to d).

The occurrence of normal, upward paleomagnetic directions appears to be confined to the basal Fortuna Formation, with reversed, downward directions found solely in the overlying Vale da Promissão Formation. The one exception to this pattern is site 19 (Fig. 2e to f), where samples of the Vale da Promissão Formation are normally magnetized close to a ca. 980 Ma mafic sill. Magnetic directions of sediments and sills in this site are similar suggesting a thermal resetting of the sediments. In consequence, site 19 was not considered in determining the mean direction of Vale da Promissão Formation. In order to test if the Vale da Promissão and Fortuna directions shared equivalent antipodal means, we have applied the McFadden and Lowes (1981) statistical test. The normal and reversed directions (Fig. 3) did not pass the reversal test suggesting that a

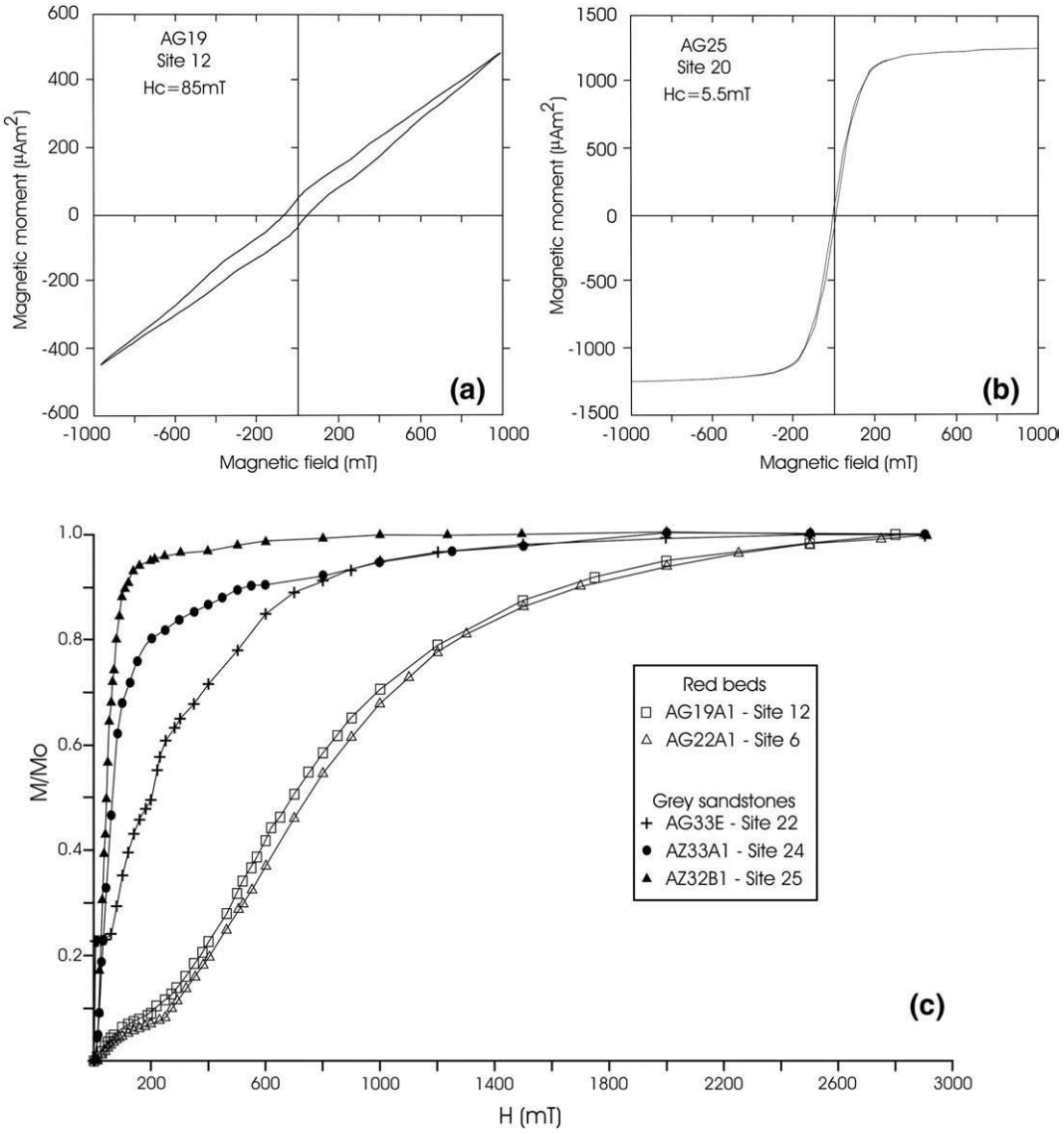


Fig. 4. Hysteresis loops obtained for samples from the Fortuna Formation (a), and Vale da Promissão Formation (b). (c) Normalized IRM-acquisition curves for red beds from the Fortuna Formation and grey sandstones form the Vale da Promissão Formation.

significant amount of geological time (and drift) has elapsed between the deposition (and magnetization) of the two formations.

3.2. Rock magnetism and textures of magnetic carriers

The higher coercivity, higher temperature component isolated from the Fortuna red beds is carried by hematite, whereas the low coercivity, random overprint observed in these rocks seems to be carried by magnetite. Hysteresis loops of Fortuna Formation rocks have a wasp-waisted form, reflecting the mixture of these high and low coercivity fractions (Fig. 4a). This bi-modal signal is also revealed by isothermal remanence (IRM) acquisition curves (Fig. 4c). They show a sharp increase in the first 200 mT, probably as a result of the tiny magnetite fraction in the samples. Then, the curves show a smooth increase in remanence, due to the hematite fraction, getting close to saturation at 3 T, the higher field produced in our pulse magnetizer.

In addition to the rock magnetic characteristics, the ubiquitous presence of hematite in Fortuna Formation samples is evident from the red color of the rocks in hand sample. In thin section, hematite occurs as massive cement in the interstitial spaces between grains, which are in clear contact with one another. This grain-supported texture suggests that the grains were not coated in iron oxides prior to deposition (Fig. 5a), but that the hematite cement filled available pore spaces during later diagenesis. This is also suggested by the intergrowth of hematite cement and "hairy" illite, a clear diagenetic feature (Fig. 5b). The phosphatic cements that overgrow detrital minerals such as monazite and zircon are also interpreted as diagenetic in origin (Fig. 5c).

Samples from the Vale da Promissão Formation show contrasting magnetic characteristics when compared to the Fortuna Formation red beds. Hysteresis loops show tight waists and fast saturation at fields below 300 mT (Fig. 4b). Samples show coercivity values between 1.1 and 24.6 mT, typical of magnetite grains. IRM-acquisition curves for some samples show a fast increase in remanence, attaining saturation at inducing fields below 300 mT (Fig. 4c). Other samples, however, show much slower increase in remanence suggesting a high coercivity fraction in addition to magnetite in these rocks (Fig. 4c). Hence, hematite may also be present in the Vale da Promissão sediments, but no consistent paleomagnetic direction is observed in these rocks above 580 °C. Reflected light microscopy of opaque, detrital minerals in polished thin sections reveals the presence of trellis-type textures that result from high temperature alteration of magmatic titanomagnetite by oxyexsolution (Haggerty, 1976).

3.3. Paleomagnetic poles

Based on the magnetic characteristics and petrographic observations, we argue that the characteristic remanent magnetizations are primary in both Fortuna and Vale da Promissão Formations. The general lack of tectonic activity in the region and the distinctiveness of these remanent directions from known Phanerozoic paleomagnetic directions support this inference. The mean directions determined from the characteristic remanences of the Fortuna Formation (normal) and the Vale da Promissão Formation (reversed) are different (Table 1), as are the magnetic mineralogies, and the ages of magnetic acquisition. We calculate two separate paleomagnetic poles for the

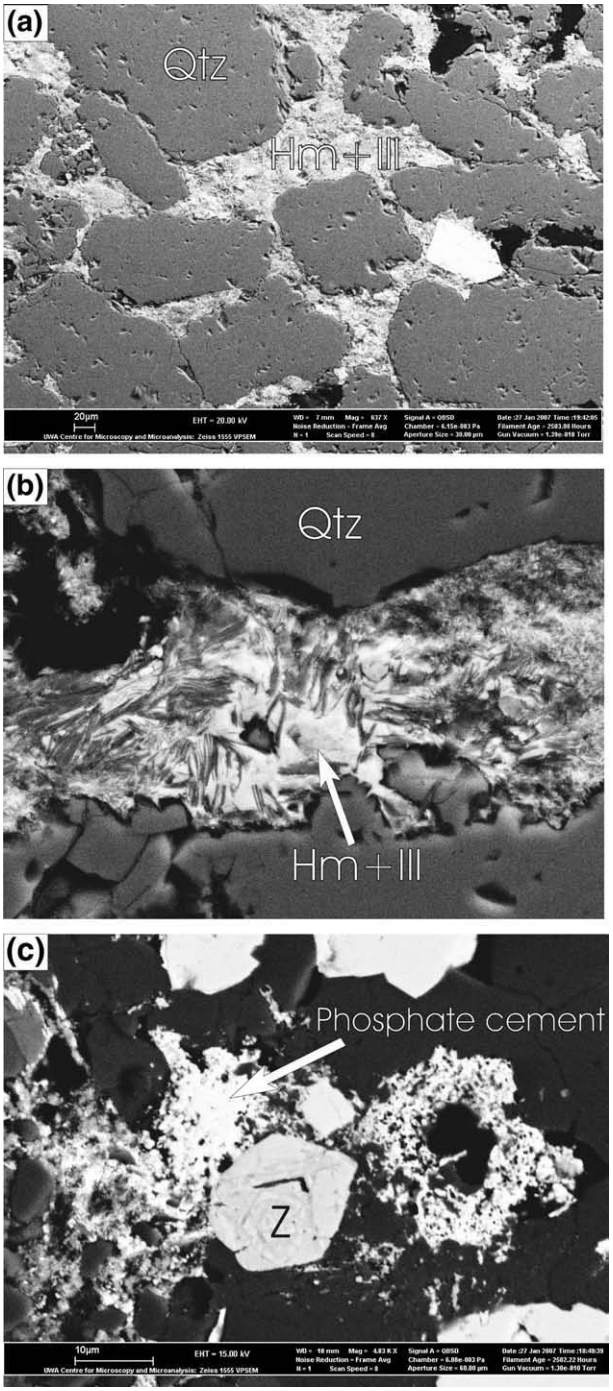


Fig. 5. a) Backscattered electron (BSE) image of Aguapeí sediments, with general view of hematite-filled pore space, b) BSE detail of hematite intergrown with "hairy" illite of authigenic origin (field of view — 50 μm) and detrital zircon, c) BSE detail of detrital zircon bordered by phosphatic cement, principally apatite.

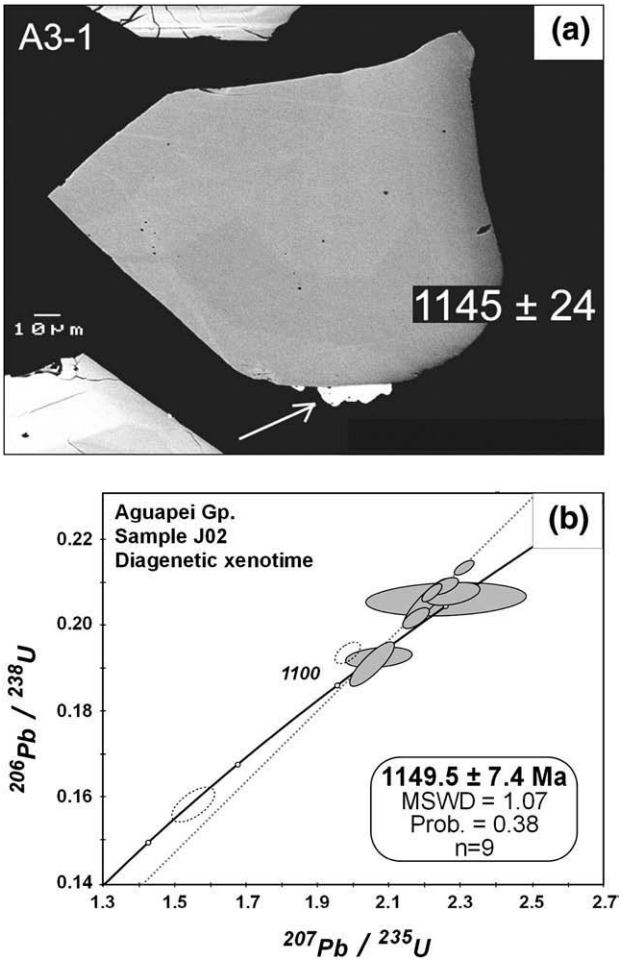


Fig. 6. (a) BSE (backscattered electrons) images of detrital zircon and xenotime overgrowths (bright white) of sample A3-1, Fortuna Formation. Arrow indicates xenotime and site of analyses. (b) U–Pb concordia diagram showing the combined results of nine SHRIMP analyses from xenotime rims from different detrital zircon grains. Discordance of the four additional results included in Table 2 plot off scale.

basal unit (FF, 59.8°N, 155.9°E, A_{95}=9.5) and the overlying formation (VdP, 49.5°N, 89.3°E, A_{95}=12.5). The FF pole is the more robust of the two, with a larger number of sites and a direct geochronological constraint (next section). The secondary hematite that is interpreted to carry the remanence from the FF basal member contrasts with the supposedly detrital origin of the magnetite in the more pelitic member of the upper VdP. If the magnetite-borne remanence is detrital in origin, the acquisition of the two paleomagnetic poles reflects the time elapsed between the sediment consolidation and diagenesis of the basal Fortuna Formation and the deposition of the overlying Vale da Promissão Formation. The age of this deposition is unknown and may be continuous with the underlying Fortuna Formation, as supposed by current workers.

3.4. Geochronology

Of the ~2500 grains examined, eighteen are marked by the presence of xenotime overgrowths (Fig. 6a) with a total of twelve grains with sufficiently thick rims to be analyzed by SHRIMP (≥10 µm). Xenotime is a rare earth-rich yttrium phosphate (YPO4) that grows as a chemical precipitate rim during diagenesis by scavenging of dissolved phosphorus from seawater or P-bearing shales (Rasmussen et al., 1998; Rasmussen et al., 2005). The SHRIMP analyses are shown in the plot of $^{206}Pb/^{238}Pb$ versus $^{207}Pb/^{235}U$ (Fig. 6b). Of the eleven analyses obtained, nine align in a regression line at 1149±7 Ma (Table 2). One age is a discordant outlier and other is a younger concordant age of 960±46 Ma. The younger and concordant age of 960 Ma is comparable to some Ar–Ar ages and may reflect a younger event, perhaps hydrothermal activity. The U and Th contents of the analyzed xenotime grains are high: 821–5531 ppm and 3602–37068 ppm, respectively. The age of 1149±7 Ma calculated from the nine remaining analyses is interpreted as the age of diagenesis and serves as a minimum age for the deposition of the Fortuna Fm., the basal member of the Aguapeí Group. Detrital zircons from other sample of Fortuna Formation (MT1) collected at São Vicente gold mine in Mato Grosso, was dated by Santos et al. (2005) where the youngest grain (d.9-1) is 1165±27 Ma old. Combining the detrital zircons information and the diagenetic xenotime data we conclude that deposition of Fortuna Formation of Aguapeí Group is constrained between 1165±27 Ma and 1149±7 Ma.

Table 2
Dados U–Pb

Spot	U	Th	Th	Pb	4f206	$^{207}Pb/^{206}Pb$	Isotopic ratios			Ages		Disc.
	(ppm)	(ppm)	(U)	(ppm)	(%)		$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{208}Pb/^{232}Th$	$^{207}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	(%)
a1.2-1	1832	15375	8.67	326	0.00	0.07763±0.71	0.2074±0.81	2.2199±1.08	0.0468±2.52	1138±14	1215±9	−6.80
a1.2-2	928	3602	4.01	154	0.00	0.07439±1.09	0.1934±1.10	1.9843±1.55	0.0431±1.36	1052±22	1140±11	−8.35
a1.4-1	1249	19597	16.22	234	5.66	0.07956±8.00	0.2060±1.52	2.2597±8.15	0.0202±2.51	1186±158	1207±17	−1.80
a1.5-1	821	5660	7.12	112	0.00	0.07108±2.28	0.1586±2.10	1.5540±3.09	0.0374±6.82	960±46	949±19	1.13
a1.6-1	1719	8726	5.25	293	3.00	0.07800±3.60	0.1925±1.00	2.0706±3.73	0.0342±1.87	1147±71	1135±10	1.04
a2.4-1	2072	18576	9.26	373	0.16	0.07813±1.24	0.2090±0.78	2.2520±1.46	0.0481±2.94	1150±25	1224±9	−6.39
a2.4-2	1916	18235	9.84	345	1.07	0.07972±2.57	0.2072±1.10	2.2776±2.79	0.0463±1.96	1190±51	1214±12	−1.99
a2.6-1	1783	37068	21.48	315	0.00	0.07790±1.19	0.2053±2.25	2.2051±2.55	0.0241±6.66	1144±24	1204±25	−5.20
a3.2-1	1226	7398	6.23	201	0.00	0.07791±1.21	0.1910±2.19	2.0522±2.50	0.0428±3.19	1145±24	1127±23	1.55
b.3-1	2967	1771	0.62	514	0.05	0.07822±1.10	0.2017±0.91	2.1756±1.42	0.0340±2.06	1153±22	1185±10	−2.78
b.3-2	5531	6692	1.25	1016	0.30	0.07855±0.72	0.2132±0.67	2.3087±0.99	0.0431±1.20	1161±14	1246±8	−7.29

U–Pb SHRIMP results from xenotime overgrowth on zircons from sample JO2, Fortuna Formation, Aguapeí Group.

4. Discussion

4.1. Age of magnetization in Fortuna and Vale da Promissão Formations

The origin of the magnetization in sedimentary rocks is a long-standing question in paleomagnetic studies (Verosub, 1977; Larson et al., 1982; Tauxe and Kent, 1984; Beck et al., 2003). The problem lies in reconciling the scale-independent magnetic properties of the bulk rock with petrographic observations that range in scale from optically-observed minerals and textures to the submicroscopic magnetic minerals or domains inferred from scanning electron or transmission electron microscopy. The origin and age of the Vale da Promissão Formation magnetization is typical of this problem. Although the presence of detrital magnetite grains is suggestive, petrographic observations alone cannot confirm that these grains host a primary, depositional remanence. Additional indication of the primary nature of magnetization in these rocks is given by an incomplete baked contact test for this unit, given by the results of site 19, sampled close to a mafic sill, compared to the other sites of the collection. Samples from sediments and the mafic intrusion collected in site 19 show concordant directions, whereas sediments collected far from any intrusion have a different direction. But even in the case where the magnetization is primary, we still face loose age constraints with the age of magnetization bracketed between the age of detrital zircons in the underlying Fortuna Formation (dated at >1165 Ma) and the age of later episode of mafic dykes intrusion (dated at 980 Ma). Although there are no significant hiatuses presently recognized in the Aguapeí Group depositional history, the difference in paleomagnetic direction and polarity between the Fortuna Formation and the overlying Vale da Promissão Formation suggests a substantial period of time elapsed between the acquisition of magnetization in these formations. The interpretation of the magnetization for the basal Fortuna Formation is simpler since the most likely carrier of the high temperature, high coercivity, characteristic remanence (ChRM) in the underlying Fortuna Formation is the abundant hematite cement. If this is correct, then the Fortuna Formation ChRM is post-depositional, since the interstitial hematite cement formed during diagenesis and lithification of these sediments. Based on the diagenetic age of 1149±7 Ma established by the U–Pb analysis of authigenic xenotime, we assign an age of magnetization for the post-depositional remanence.

4.2. Implications for Rodinia

Proposed permutations of the Rodinia supercontinent began to arise soon after the original geometries of Hoffman (1991) and Dalziel (1992) were advanced. The later, derivative models mostly adhere to the key assumption of a Laurentia-centered supercontinent that was assembled via Grenvillian collisions on the (present-day) eastern margin of proto-North America and later disassembled during Neoproterozoic rifting. The paleogeography of the intervening ~500 Ma initially attracted little comment, other than some debate as to when rifting along the (present-day) western margin of Laurentia began (Prave, 1999). This apparent consensus has been undermined by the recent work of both geologists [e.g., (Hanson et al., 2004; Kroner and Cordani, 2003; Cordani et al., 2003)] and paleomagnetists [e.g., (Wingate et al., 2002; Pisarevsky and Natapov, 2003; Meert and Torsvik, 2003; Tohver et al., 2006)] questioning the monolithic nature of the Rodinia supercontinent, in particular the assertion that all of the continents were arrayed near or alongside Laurentia. For example, the elegant simplicity of the original match in Grenvillian-aged metamorphic provinces has been clouded by the recognition that cooling ages are preserved throughout the Laurentian belt, whereas the southwestern Amazon retains a mixture of cooling and deformation ages (Tohver et al., 2005a,b). In the last few years an evolving Laurentia–Amazon connection has been proposed to explain structural, isotopic and geophysical data on the southwestern Amazon and eastern Laurentia (Tohver et al., 2004a,b, 2005b). The new paleomagnetic data that we present here do support a direct, tectonic link between Laurentia and Amazonia at 1.15 Ga, but they suggest a dynamic Rodinia that contrasts with the stasis of the original models.

The Fortuna Formation provides a well-dated, well-characterized paleomagnetic pole based on a direction that is distinct from any known overprinting directions. Using the quality criteria of Van der Voo (1990), the pole is assigned a Q-factor of 5. Given the uncertainties outlined above regarding the Vale da Promissão Formation ChRM, we restrict the following discussion of paleogeography to the 1.15 Ga position established for the Amazon craton by the Fortuna Formation. Considering the new pole presented here and the Nova Floresta pole (NF) presented by Tohver et al. (2002) a segment of apparent polar wander path (APWP) for Amazonia can be traced between 1200 and 1149 Ma. In case there was no relative motion between Amazonia and Laurentia, this APWP can be compared to that of Laurentia. As a first approximation we consider the Laurentian path defined by paleomagnetic poles from Upper Bylot (UB, 1204±22 Ma, 8°N, 205°E, A_{95}=4) and Abitiby dikes (AD, 1141±1 Ma, 43°N, 208°E, A_{95}=14). A rotation of 41.9° around an Euler pole at 60.35°N and 310.7°E would account for the match of the two APWP segments defined above. But when applied to the continents, this rotation places the western margin of Amazonia more than 5000 km far from the eastern margin of Laurentia (Fig. 7). Given the abundant evidence for collision in both cratons in this time interval this paleogeographic configuration would require a significant intervening block between Laurentia and Amazonia by this time. Moreover, isotopic evidence for Amazonian heritage in blocks of the present-day eastern margin of Laurentia and vice-versa argue strongly on the connection between these two cratons.

We prefer instead to compare pole by pole the Amazonia and Laurentia database. According to this approach, the relative position of the Amazon craton determined from the Nova Floresta and Fortuna Formation poles is depicted in Fig. 7. In these reconstructions, the Amazon craton has shifted alongstrike from the original 1.2 Ga collision site farther southeast (present-day), typified by the Llano Uplift region, toward the (present-day) SE Appalachians. Geological evidence for this

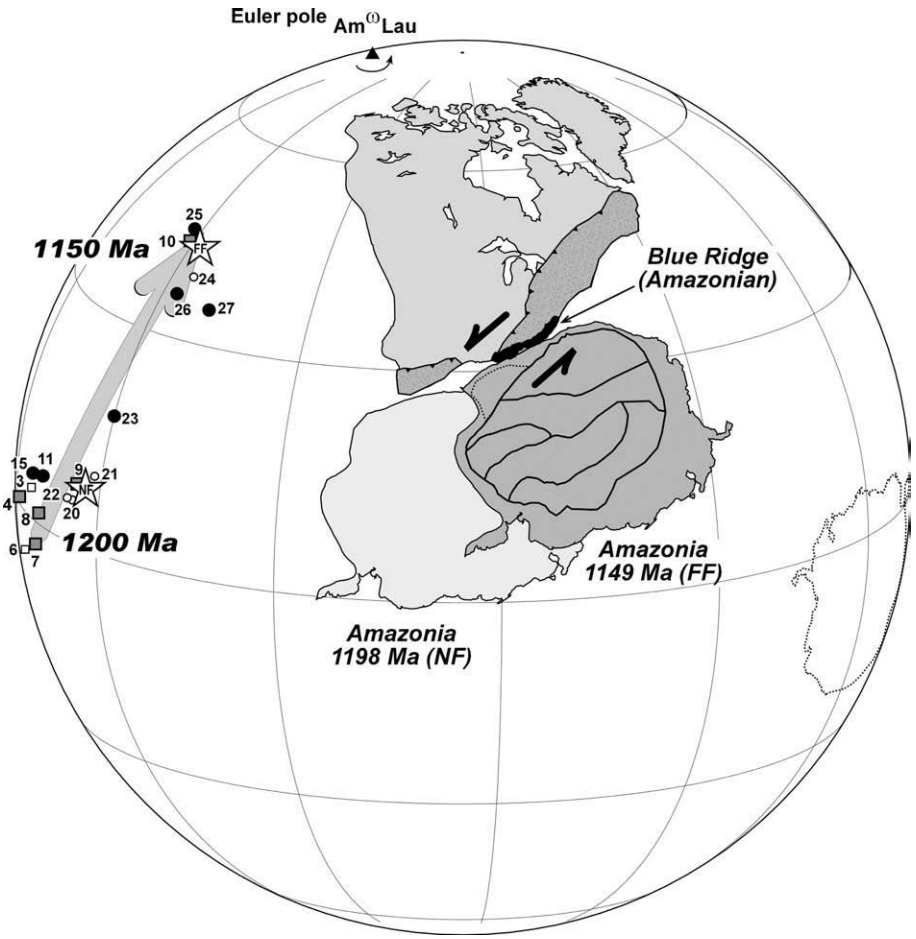


Fig. 7. The along-strike migration of the Amazon craton relative to Laurentia indicated by the ~2000 km shift from the 1.2 Ga position (Tohver et al., 2002) to the 1.15 Ga position using the Fortuna Formation paleomagnetic pole (Euler pole 10.5°N, 298.7°E, −150°). Grey arrow indicates the APW path for proto-North America. Individual poles for Greenland (circles) and Laurentia (squares) are numbered according to Tohver et al. (2002). Stars indicate Amazonia poles (NF: Nova Floresta, FF: Fortuna Formation) rotated according to each reconstruction. The Euler rotation pole (triangle — 75°N, 155.5°E, 15°) that takes the Amazon craton from 1.2 Ga to 1.15 Ga is also shown. Matching the APWPs, as discussed in the text, results in an alternative position for the Amazon craton (dashed outline) with no Grenvillian connection between Amazonia and Laurentia.

motion is found in the presence of large-scale, sinistral strike-slip shear zones active throughout the 1.2–1.0 Ga interval in SW Amazonia (Tohver et al., 2005b; Litherland et al., 1989). The 1.15 Ga position of the Amazon craton opposite the present-day SE Appalachians has left other telltale signs, notably the rocks of the western Blue Ridge/Mars Hills (BRMH) terrane which are exotic to Laurentia. A non-Laurentian mantle reservoir for the BRMH rocks was identified using Pb and Nd isotopic signatures (Loewy et al., 2003; Tohver et al., 2004a), and rock samples from this terrane contain relict zircons with ages characteristic of SW Amazonia (Carrigan et al., 2003). The age of the high-grade deformation in the SE Appalachians that attended the docking of these rocks onto the Grenvillian margin of Laurentia is constrained to the 1145–1075 Ma interval (Aleinikoff et al., 2000). Isotopic evidence points to docking during the earlier portion of this 70 Ma interval, since the ε_{Nd} values of rocks much younger than 1.15 Ga are positive, reflecting a Laurentian mantle source (Tohver et al., 2004a). We also note that the NE–SW directed compressional event discerned by Aleinikoff et al. (2000) for this deformation is parallel to the (present-day) strike of Grenvillian rocks, echoing the earlier findings of Mosher (1998) that the colliding continent approached from the (present-day) SW. Thus, the NE directed compression parallel to the Llano and Grenville tectonic fronts is compatible with a strike-slip scenario, observations that are comfortably accounted for by the ongoing strike-slip motion of Amazonia.

This dynamic scenario involves a net sinistral strike-slip motion of the Amazon craton across Laurentia's margin between 1.2 and 1.0 Ga, initiating with the collision along the southernmost Laurentia (Llano region, Texas), reaching the SE Appalachians by 1.15 Ga. This motion is described by a 15° rotation about Euler pole located at 75°N, 155.5°E (Fig. 7), with the general NE–SW trend of the Grenvillian margin approximating a small circle to this Euler pole. Continued motion along this small circle within an overall transpressive setting would result in net strike-slip motion and contribute to the long span of Grenvillian orogenesis (1.2–1.0 Ga).

In terms of Amazon geology, the opening of the Nova Brasilândia rift after 1220 Ma (youngest detrital zircon age; (Santos et al., 2001)) and closing by 1100 Ma, the age of peak metamorphism (Tohver et al., 2004b; England and Thompson,

1984), could have been a consequence of the 1.2 Ga Grenvillian collision between "greater" Amazonia and Laurentia. This interpretation observes the temporal correlation of the Nova Brasilândia rift sediments and the Aguapeí Group sediments suggested by Santos et al. (2001, 2005). Intriguingly, this impactogenic origin of the Nova Brasilândia rift is similar to that ascribed to the contemporary 1.1 Ga Keweenawan and Umkondo rifts. These latter rifts are synchronous with "Grenvillian" deformation of the Laurentian and Kalahari cratons, respectively (Hanson et al., 2004), but were spared the subsequent closure and high-grade metamorphism that affected the Nova Brasilândia rift. The position of Amazonia alongside Laurentia at the time of the Umkondo and Keweenawan rifting supports the contention of Hanson et al. (2004) that the occurrence of synchronous, rift-related volcanism in interior of the Kalahari and Laurentian cratons is purely coincidental, not indicative of a common paleogeography.

5. Conclusions

The 1.15 Ga position of the Amazon craton indicated by the Fortuna Formation is based on a stable, well-characterized magnetization (18 sites, 115 samples) hosted by abundant hematite cement. The age of diagenesis of these rocks is established by the 1149±7 Ma U–Pb age of authigenic xenotime that occurs as rims on detrital zircon. The second pole reported in this study from the overlying mudstones of the Vale da Promissão Formation is less well-constrained in age. Using Fortuna Formation as a control on the position of the Amazon craton implies that the closure of the Nova Brasilândia rift occurred ~50 Ma before the peak metamorphism recorded by the high-grade metasediments and metabasic rocks in that inverted rift. Strike-slip motion following the original 1.2 Ga caused the Amazon craton to migrate ~2000 km to a position near the (present-day) SE Appalachians. This position corroborates the identification of Amazonian crust in the SE Appalachian basement. Taken together, these new data suggest a dynamic paleogeography for the Rodinia supercontinent in its assembly phase.

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References

Aleinikoff, J.N., Burton, W.C., Lyttle, P.T., Nelson, A.E., Southworth, C.S., 2000. U–Pb geochronology of zircon and monazite from Mesoproterozoic granitic gneisses of the northern Blue Ridge, Virginia and Maryland, USA. Precambrian Res. 99, 113–146.

Beck, M.E., Burmester, R.F., Housen, B.A., 2003. The red bed controversy: shape analysis of Colorado Plateau units suggests long magnetization times. Tectonophysics 362, 335–344.

Boger, S.D., Raetz, M., Giles, D., Etchart, E., Fanning, C.M., 2005. U–Pb age data from the Sunsas region of Eastern Bolivia, evidence for the allochthonous origin of the Paragua Block. Precambrian Res. 139, 121–146.

Bond, G.C., Nickeson, P.A., Komins, M.A., 1984. Breakup of a supercontinent between 625 and 555 Ma: new evidence and implications for continental histories. Earth Planet. Sci. Lett. 70, 325–345.

Carrigan, C.W., Miller, C.F., Fullagar, P.D., Bream, B.R., Hatcher, R.D., Coath, C.D., 2003. Ion microprobe age and geochemistry of southern Appalachian basement, with implications for Proterozoic and Paleozoic reconstructions. Precambrian Res. 120, 1–36.

Cordani, U.G., D'Agrella-Filho, M.S., Brito-Neves, B.B., Trindade, R.I.F., 2003. Tearing up Rodinia: the Neoproterozoic palaeogeography of South American cratonic fragments. Terra Nova 15, 350–359.

Dalziel, I.W.D., 1992. On the organization of American plates in the Neoproterozoic and the breakout of Laurentia. GSA Today 2 (11), 237–241.

Donnadieu, Y., Goddéris, Y., Ramstein, G., Nédélec, A., Meert, J., 2004. Continental breakup and global climatic evolution. Nature 428, 303–306. doi:10.1038/nature02408.

England, P.C., Thompson, A.B., 1984. Pressure–temperature–time paths of regional metamorphism, part I, heat transfer during the evolution of regions of thickened continental crust. J. petrol. 25, 894–928.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. Roy Soc. London A217, 295–305.

Fletcher, I.R., McNaughton, N.J., Aleinikoff, J.A., Rasmussen, B., Kamo, S., 2004. Improved calibration procedures and new standards for U–Pb and Th–Pb dating of Phanerozoic xenotime by ion microprobe. Chem. Geol. 209, 295–314.

Geraldes, M.C., Figueiredo, B.R., Tassinari, C.C.G., Ebert, H.D., 1997. Middle Proterozoic vein-hosted gold deposits in the Pontes e Lacerda region, southwestern Amazonian Craton, Brazil. Int. Geol. Rev. 39, 438–448.

Geraldes, M.C., Van Schmus, W.R., Condie, K.C., Bell, S., Teixeira, W., Babinski, M., 2001. Proterozoic geologic evolution of the SW part of the Amazonian Craton in Mato Grosso state, Brazil. Precambrian Res. 111, 91–128.

Haggerty, S.E., 1976. Oxidation of opaque mineral oxides in basalts, Reviews in Mineralogy. Min. Soc. Am. 3, 1–100.

Hanson, R.E., Crowley, J.L., Bowring, S.A., Ramezani, J., Gose, W.A., Dalziel, I.W.D., Pancake, J.A., Seidel, E.K., Blenkinsop, T.G., Mukwakwami, J., 2004. Coeval large-scale magmatism in the Kalahari and Laurentia cratons during Rodinia assembly. Science 304, 1126–1129.

Hodges, K.V., 2000. Tectonics of the Himalaya and southern Tibet from two perspectives. Geol. Soc. Amer. Bull. 112, 324–350.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside–out? Science 252, 1409–1412.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Kroner, A., Cordani, U., 2003. African, southern Indian and South American cratons were not part of the Rodinia supercontinent: evidence from field relationships and geochronology. Tectonophysics 375, 325–352.

Larson, E., Walker, T., Patterson, P., Hoblitt, R., Rosenbaum, J., 1982. Paleomagnetism of the Moenkopi Formation, Colorado Plateau: basis for long-term model of acquisition of chemical remanent magnetism in red beds. J. Geophys. Res. 87, 1081–1106.

Leite, J.A.D., Saes, G.S., 2003. Geocronologia Pb/Pb de zircões detríticos e análise estratigráfica das coberturas sedimentares Proterozóicas do sudeste do Craton Amazônico. Revista do Instituto de Geociências, Geol. USP Ser. Cient., São Paulo 3, 113–127.

Litherland, M., Annells, R.N., Darbyshire, D.P.F., Fletcher, C.J.N., Hawkins, M.P., Klinck, B.A., Mitchell, W.I., O'Conner, E.A., Pitfield, P.E.J., Power, G., Webb, B.C., 1989. The Proterozoic of Eastern Bolivia and its relationship to the Andean mobile belt. Precambrian Res. 43, 157–174.

Loewy, S.L., Connelly, J.N., Dalziel, I.W.D., Gower, C.F., 2003. Eastern Laurentia in Rodinia: constraints from whole-rock Pb and U/Pb geochronology. Tectonophysics 375, 169–197.

McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J. R. Astron. Soc. 67, 19–33.

Meert, J.G., Torsvik, T.H., 2003. The making and unmaking of a supercontinent: Rodinia revisited. Tectonophysics 375, 261–288.

Mosher, S., 1998. Tectonic evolution of the southern Laurentian Grenville orogenic belt, Geol Soc. Am. Bull. 110, p.1357–1375.

Pinho, M.A.S.B., Chemale Jr, F., Van Schmus, W.R., Pinho, F.E.C., 2003. U–Pb and Sm–Nd evidence for 1.76–1.77 Ga magmatism in the Moriru region, Mato Grosso, Brazil: implications for province boundaries in the SW Amazon Craton. Precambrian Res. 126, 1–25.

Pisarevsky, S.A., Natapov, L.M., 2003. Siberia and Rodinia. Tectonophysics 375, 221–245.

Prave, A.R., 1999. Two diamictites, two cap carbonates, two δ 13C excursions, two rifts: the Neoproterozoic Kingston Peak Formation, Death Valley, California. Geology 27, 339–342. doi:10.1130/0091-7613 (1999)027<0339:TDTCCT>2.3.CO;2.

Rasmussen, B., Blake, T.S., Fletcher, I.R., 2005. U–Pb zircon age constraints on the Hamersley spherule beds: Evidence for a single 2.63 Ga Jeerinah–Carawine impact ejecta layer. Geology 33, 773–776.

Rasmussen, B., Fletcher, I.R., Sheppard, S., 1998. Isotopic dating of the migration of a low grade metamorphic front during orogenisis. Earth Planet. Sci. Lett. 164, 135–149.

Rizzotto, G.J., 1999. Petrologia e geotectônica do Grupo Nova Brasilândia, Rondônia, MSc Thesis. 131pp., Uni. Fed. Rio Grande do Sul, Porto Alegre, Brazil.

Ruiz, A.S., 2005. Evolução geológica do Sudoeste do Craton Amazônico, Região limítrofe Brasil–Bolívia. PhD Thesis, Rio Claro, São Paulo, Brazil. pp. 258.

Saes, G.S., Leite, J.D., 1993. Evolução tectono-sedimentar do Grupo Aguapeí, Proterozoico Medio, na porção meridional do craton Amazônico: Mato Grosso e Oriente Boliviano. Revista Brasileira de Geociências 23, 31–37.

Santos, J.O.S., Hartmann, L.A., Gaudette, H.E., Groves, D.I., McNaughton, N.J., Fletcher, I.R., 2000. A new understanding of the provinces of the Amazon Craton based on integration of field mapping and U–Pb and Sm–Nd geochronology. Gondwana Res. 3, 453–488.

Santos, J.O.S., McNaughton, N.J., Hartmann, L.A., Fletcher, I.R., Salinas, R.M., 2005. The age of deposition of the Aguapeí Group, Western Amazon Craton, based on U–Pb study of diagenetic xenotime and detrital zircon. Congreso Latinoamericano de Geologia, 12. Quito, Ecuador May Ministerio de Energia y Minas. Area Técnica: Geologia, group 9. ISBN 9978-44-206-5. CD-ROM.

Santos, J.O.S., Rizzotto, G.A., Hartmann, L.A., McNaughton, N.J., Fletcher, L.R., 2001. Ages of sedimentary basins related to the Sunsas and Juruena orogenies, Southwest Amazon Craton established by zircon U–Pb geochronology. Workshop Geology of the SW Amazonian Craton: State-of-the-Art. São Paulo, Extended abstracts, pp. 114–118.

Souza, E.P., Hildred, P.R., 1980. Contribuição ao estudo da geologia do Grupo Aguapeí, oeste de Mato Grosso, In: 31° Congresso Brasileiro de Geologia. Camboriú, Anais 2, 813–825.

Tassinari, C.C.G., Bettencourt, J.S., Geraldes, M.C., Macambira, M.J.B., Lafon, J.M., 2000. The Amazonian Craton. In: Cordani, U.G., Milani, E.J., Thomaz-Filho, A., Campos, D.A. (Eds.), Tectonic Evolution of South America, pp. 41–95. Rio de Janeiro 31st International Geological Congress, 2000.

Tauxe, L., Kent, D.V., 1984. Properties of a detrital remanence carried by hematite from a study of modern river deposits and laboratory redeposition experiments. Geophys. J. R. Astron. Soc. 77, 543–561.

Thompson, A.B., Schulmann, K., Jezek, J., 1997. Thermal evolution and exhumation in obliquely convergent (transpressive) orogens. Tectonophysics 280, 171–184.

Tohver, E., van der Pluijm, B.A., Van der Voo, R., Rizzotto, G., Scandolara, J.E., 2002. Paleogeography of the Amazon craton at 1.2 Ga: early Grenvillian collision with the Llano segment of Laurentia. Earth Planet. Sci. Lett. 199, 185–200.

Tohver, E., Bettencourt, J.S., Tosdal, R., Mezger, K., Leite, W.B., Payolla, B.L., 2004a. Terrane transfer during the Grenville orogeny: tracing the Amazonian ancestry of southern Appalachian basement through Pb and Nd isotopes. Earth Planet. Sci. Lett. 228, 161–176 C.C.G.

Tohver, E., van der Pluijm, B., Mezger, K., Essene, E., Scandolara, J., Rizzotto, G., 2004b. Significance of the Nova Brasilândia metasedimentary belt in western Brazil: redifining the Mesoproterozoic boundary of the Amazon craton. Tectonics 23, TC6004. doi:10.1029/2003TC001563.

Tohver, E., Teixeira, W., van der Pluijm, B., Geraldes, M.C., Bettencourt, J.S., Rizzotto, G., 2005a. Restored transect across the exhumed Grenville orogen of Laurentia and Amazonia, with implications for crustal architecture. Geology 34 (8), 669–672. doi:10.1130/G22534.1.

Tohver, E., van der Pluijm, B.A., Scandolara, J.E., Essene, E.J., 2005b. Late Mesoproterozoic deformation of SW Amazonia (Rondônia, Brazil): geochronological and structural evidence for collision with southern Laurentia. J. Geology 113, 309–323.

Tohver, E., D'Agrella-Filho, M.S., Trindade, R.I.F., 2006. Paleomagnetic record of Africa and South America for the 1200–500 Ma interval, and evaluation of Rodinia and Gondwana assemblies. Precambrian Res. 147, 193–222.

Torsvik, T.H., 2003. The Rodinia jigsaw puzzle. Science 300, 1379–1381.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.

Verosub, K.L., 1977. Depositional and post-depositional processes in the magnetization of sediments. Rev. Geophys. Space Phys. 15, 129–143.

Wingate, M.T.D., Pisarevsky, S.A., Evans, D.A.D., 2002. Rodinia connections between Australia and Laurentia: no SWEAT, no AUSWUS? Terra Nova 14, 121–128.